

Multiple Choice (10 marks)

1. Pat is learning how energy flows through an ecosystem. Pat understands that animals get energy from plants. What do plants do that allow animals to get energy from them?
 - (a) Plants keep water in their roots.
 - (b) Plants release nitrogen into the soil.
 - (c) Plants provide shelter from the weather.
 - (d) Plants make sugar in the presence of light.
2. Which of these is an example of an adaptation for defense from predators?
 - (a) giraffe's long neck
 - (b) baby deer's spots
 - (c) otter's oily fur
 - (d) fish's gills
3. Which planet in the solar system has the longest year?
 - (a) The planet closest to the Sun.
 - (b) The planet with the longest day.
 - (c) The planet with the most moons.
 - (d) The planet farthest from the Sun.
4. A student poured water into a plastic tray. The student then put the tray into the freezer. What property of the water changed as the water froze?
 - (a) The water became a gas.
 - (b) The mass of water increased.
 - (c) The water took a definite shape.
 - (d) The flavor of the water changed.
5. Rossby waves are slow-moving, long ocean waves. What feature of a Rossby wave would change if the wavelength was doubled?
 - (a) The amplitude would double.
 - (b) The frequency would double.
 - (c) The amplitude would decrease by half.
 - (d) The frequency would decrease by half.

True / False (10 marks)

Answer True or False

6. True or False: A major volcanic eruption can release ash and sulfur gas, leading to unusual climate conditions like cold and acidic rain.
7. True or False: A tornado can create a narrow path of destruction through a forest due to its intense winds.
8. True or False: Evaluating and interpreting data is essential for drawing valid conclusions in scientific investigations.
9. True or False: A metal can decays faster than an apple core because it is made from natural materials.
10. True or False: Alternative fuels are widely available at every gas station, making them a common choice for drivers.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the most effective method for measuring the size of oak tree leaves, including the importance of labeling in your explanation.
12. Discuss the most effective way to visually represent the percentage of minerals in granite samples collected from the mountains of western Nevada, and explain your reasoning.

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